HOSAM IBRAHIM

Engineering Leader | Senior Solutions Architect | Senior Software Engineer
AML/CFT Financial Intelligence | Systems & Data Integration
Cairo, Egypt | hosam.fazie@hotmail.com
linkedin.com/in/hosam-ibrahim | hosam-f-ibrahim.github.io

PROFILE

Engineering leader and solutions architect with 20+ years of professional software experience across financial intelligence, enterprise operations and industrial automation. Translates business and information needs into requirements, technical direction and software solutions, working with stakeholders through business and systems analysis. Grounds architecture in practical web, database and service development, helping teams adapt interconnected software through integration, legacy-data migration and reusable, extensible design. Has led engineering teams, including feedback, performance assessment, career development and hiring for engineers in the reporting line. Most recently, as head of engineering, set technical direction and engineering priorities, and oversaw delivery. Client-facing work spans discovery, solution design, proposals and scoping. Applies engineering judgment to keep AI-assisted development aligned with product needs and architectural decisions.

PROFESSIONAL EXPERIENCE

Project periods are shown separately from role periods; a project is not attributed to a later role.

intelliSYS | intellisys.sk — *October 2019–September 2026*

Head of Engineering — *January–September 2026*

- Led the engineering team, including hiring and coaching; set technical direction and architecture; and stayed hands-on in development and technical problem-solving.
- Oversaw delivery, quality and engineering processes; set engineering priorities, resources and budget; and worked with clients and business stakeholders on solutions.
- Mentored and assessed engineers and ran technical hiring interviews.
- Guided architecture and product direction for Learning Clarity; FIU360 remained within this role's responsibilities.
- Carried Senior Solutions Architect responsibilities alongside this formal role.

Head of Professional Services / Senior Solutions Architect — *April 2020–January 2026*

- Led technical discovery and solution design with prospective or existing clients before a contract or scope was agreed.
- Led and delivered client demos and proof-of-concept work across different engagements; wrote or substantially shaped technical proposals, RFP responses and RFI responses; and prepared effort estimates and technical scope/SOW content.
- Designed add-on and plugin extensions that closed business gaps in existing systems, and data-transformation solutions for migrating legacy systems to new platforms.
- On-site and remote client work across Europe, Africa and the Middle East.

Software Engineering Team Lead — *December 2019–March 2020*

- Led the engineering team through requirements analysis, system design, and data transformation, migration and integration solutions.

Senior Software Engineer — *October–December 2019*

- Hands-on design and development with C#/.NET, Microsoft SQL Server, REST APIs and web services, and ETL; promoted to Software Engineering Team Lead in December 2019.

intelliSYS project records (main projects first)

FIU360 | intellisys.sk/fiu-360 — *March 2020–September 2026*

- Led the architecture, design and delivery of FIU360, a scalable, modular AML/CFT financial intelligence platform meeting AML/CFT standards (FATF, UN) and covering a national financial intelligence unit's full operating cycle, from incoming reports to intelligence disseminated to law enforcement agencies.
- Registration with role-based access; data collection through a flexible XML engine (300+ parameters) for CTR, STR, EFT and AIF reports, with end-to-end report intake of over 100K records/hour on a commodity server; sanctions screening against watchlists with alerts, case confirmation, whitelisting and auto-escalation.
- Data enrichment (FIU360 Connect) from financial, national/international and open sources; AI-driven search and advanced indexing; AI-powered analytics using LLMs for real-time operational intelligence and strategic insights.

intelliSYS (continued) | October 2019–September 2026

- Interactive data and relationship visualization; case management with a specialized AML/CFT module, improving closure rates; document management with full-text search; one-click FATF-compliant statistics reports; automated dissemination that freed analyst time; drag-and-drop BI dashboards.
- Championed Agile delivery and CI/CD in Azure DevOps, and mentored cross-functional teams in full-stack architecture, data engineering, AI integration and regulatory strategy.

  Technologies: C#, .NET and .NET Core; REST/Web APIs; Microsoft SQL Server (T-SQL, views, stored procedures, jobs, indexes); XML processing (XML Schema, XPath, XQuery); Vue.js; DevExpress; Azure DevOps

Data Enrichment | intellisys.sk/additional-tools *October 2019–January 2020*

- Architected an extensible .NET/C# add-on suite that automated enrichment of AML reports with data from financial, government and open sources, removing manual data gathering and increasing data security and accuracy.
- Built ETL pipelines with 99% data accuracy; used multithreading to collect data from multiple sources in parallel.
- Automated data acquisition on FIU report receipt or analyst request, cutting latency by 99%; validation, error handling and retry logic lowered enrichment failures by more than 90%.
- Secured data in transit with TLS encryption, monitored health with Application Insights, and worked with third-party API providers on data coverage and SLA compliance.
- Later continued hands-on design and development on the add-on (FIU360 Connect), which extends FIU360 and other existing FIU systems without changing their code base.

  Technologies: C#/.NET; Microsoft SQL Server; REST APIs and web services; ETL

Data Visualization | intellisys.sk/additional-tools *January 2020–July 2021*

- Architected and led development of a scalable .NET/C# add-on framework that renders all database objects and relationships in FIU360 and other existing FIU systems without changing their code base; together with data enrichment, it cut analyst effort by 90%.
- Built interactive dashboards with drill-down and filters, dynamic connectors for cross-database relationship mapping, REST services for real-time data, and graphing components for complex entity networks.
- Optimized SQL Server views and stored procedures to pre-aggregate relationship data, cutting load times by 50%, and built parallel ETL pipelines.
- Established CI/CD pipelines in Azure DevOps with zero-downtime deployments, monitored performance with Application Insights, ran requirements workshops and mentored junior developers.

  Technologies: C#/.NET; Microsoft SQL Server; REST APIs and web services; DevExpress

Learning Clarity | learningclarity.ai *January–September 2026*

- Multilingual AI learning platform for web, iOS and Android that turns learners' notes, PDFs and documents into lessons, quizzes, timed exams and podcasts.
- Guided architecture and product direction for the team building Learning Clarity.
- Live AI features covered prompts, model calls and document retrieval for AI answers: for some, directed a coding AI on the AI logic and reviewed its code or configuration; for others, chose the AI design and reviewed outcomes while the team handled implementation and testing.

  Technologies: TypeScript web app; Supabase backend (Postgres, authentication, edge functions, cron jobs); Flutter mobile apps for Android and iOS

AML PRO | intellisys.sk/aml-pro *January–December 2022*

- Reporting application used by reporting entities' compliance officers.
- Implemented a .NET/C# application for FIU XML reporting, enabling reporting entities to submit compliant data from day one and reducing onboarding time.
- Built an External Data Manager to import and normalize client data from Excel and organizational databases, and a Client Data Manager whose validation rules flagged and corrected inconsistencies before submission; DevExpress UI for compliance officers.
- Worked with compliance, legal and business stakeholders on requirements; mentored junior engineers on ETL design, data validation and .NET development, raising team productivity; enforced pull-request reviews and branching strategies in Git.

  Technologies: C#/.NET; Microsoft SQL Server; XML Schema; DevExpress

Rakaaiz

September 2014–September 2019

Business/System Analyst & Lead Software Developer *September 2014–September 2019*

- Business and systems analysis, solution design and development of .NET and SQL Server business applications for clients in the UAE, across event management, finance, training, HR and nonprofit operations.
- Architected, designed and developed further solutions for real estate, geolocation, meeting management and correspondence management.

Rakaaiz project records (project period for each record below: September 2014–September 2019)

Event Management System

- Led full lifecycle development of a .NET and SQL Server event-management system covering planning, registration, payment processing, attendee communication and reporting.

Financial Management System

- Led development of a .NET financial-management system with budgeting, forecasting and general-ledger modules, data validation and enterprise web-service integration, used by finance teams for financial planning, analysis and reporting.

  Technologies: .NET; Microsoft SQL Server; Entity Framework; AngularJS; JavaScript; web services

Training Management System

- Engineered a .NET and SQL Server training and certification system that let each organization design its own certificates, with digital signatures for authenticity and integrity.

HR Management System

- Engineered a .NET and SQL Server HR system for employee records, payroll and benefits administration, with secure authentication and workforce reporting.

NPO Operation Management System

- Developed a modular .NET and SQL Server system for nonprofit operations, including donor management and program tracking, with encryption of sensitive data.

Freelance

January 2014–September 2019

Software Engineer *January 2014–September 2019*

- Delivered end-to-end software solutions for clients, from business, systems and user-requirements analysis to high-level solution design and low-level algorithm design.
- Led and planned projects; from September 2014, continued freelance work alongside Rakaaiz.

Appointment Reservation System

- Developed a C#/.NET and SQL Server appointment and time-slot management system for a client in Egypt, enabling clinics, labs and other businesses to manage online booking, rescheduling and cancellation.

Social Networking Platform

- Contributed to a .NET and SQL Server social-networking platform for a client in Canada, including user profiles, content sharing and social-interaction features.

ASEC Automation

November 2009–December 2013

Software Engineer *November 2009–December 2013*

- Owned the full development lifecycle of .NET and ASP.NET applications for industrial automation and enterprise operations, from design to implementation.

ASEC Automation project records (project period for each record below: January 2010–December 2013)

OPC Client Software

- Developed .NET OPC client software for real-time data acquisition and control with process-automation servers, with an architecture for continuous data streams and real-time visualization for operators.

Time Attendance Tracking System

- Developed an ASP.NET and SQL Server time-attendance system for clock-in and clock-out, time-off requests and payroll reports, integrated with HR and payroll systems.

Financial Planning System

- Developed a financial planning and budgeting system using Microsoft Access VBA and Excel, with a calculation engine for projections and variance analysis, and Crystal Reports reporting.

Freelance

January–November 2009

Development Team Leader *January–November 2009*

- Led a development team delivering client software, from business, systems and user-requirements analysis through high-level solution design and low-level algorithm design.
- Handled project planning and scheduling.

Travel Agency Management System

- Engineered a .NET and SQL Server travel-agency system for bookings, customer records, travel packages and invoicing.

DPME

January 2006–December 2008

Technical Lead *January 2006–December 2008*

- Business, systems and user-requirements analysis; high-level solution design; team leading and training; project planning and scheduling.

DPME ERP *January 2006–December 2008*

- Led a full lifecycle development team through design, implementation and migration of an ERP system covering project management, finance, HR, payroll and timesheets; contributed architecture and development using C#, ASP.NET and SQL Server.
- Automated critical business workflows, reducing administrative overhead by more than 50%, and migrated legacy data into the new system while keeping data integrity.

SKILLS

Architecture and leadership: Software Architecture; Solution Architecture; Team Leadership; Leading Development Teams; System Integration; Requirements Analysis; Business Analysis; Systems Analysis; Project Planning; Coaching & Mentoring

Data and integration (project-associated): Data Integration; Data Transformation; Extract, Transform, Load (ETL); Data Visualization; XML Data Processing; Data Validation; Data Migration

Tools: Azure DevOps; Git; Application Insights

Historical technologies: OPC (OLE for Process Control); Industrial Automation; AngularJS; Windows Communication Foundation (WCF)

Course and credential exposure: Artificial Intelligence (AI); Machine Learning; Amazon Web Services (AWS); Python (see Certifications and Courses)

EDUCATION

BSc in Commerce (Accounting) | Cairo University *1997–2002*

CERTIFICATIONS AND COURSES

Certificate of Completion – Elements of AI | University of Helsinki *September 2025*
Python Programming | Kaggle *September 2025*
AWS Educate Machine Learning Foundations | AWS *July 2025*
Introducing Generative AI with AWS | Udacity *June 2025*
MCTS .NET Framework 2.0 Distributed Applications | Microsoft *2007*
MCTS SQL Server 2005 | Microsoft *2007*
MCPD Enterprise Applications | Microsoft *2007*

LANGUAGES

Arabic — native/bilingual | English — professional working proficiency